

Prompt	Continuation (== 6B)	2.7B	1.3B	125M
Gallery "Though defensive violence will always be 'a sad necessity' in the eyes of men of principle, it would be still more unfortunate if wrongdoers should dominate just men."- St. Augustine "A new idea is first	condemned as ridiculous, and then dismissed as trivial, until finally it becomes what everybody knows." - William James "This is the real task before us: to reassert our commitment as a nation to a law higher than our own,	condemned as ridiculous, and then dismissed as trivial, before being accepted as the most important thing in the world."- George Bernard Shaw "The ultimate measure of a man is not where he stands in moments of comfort and convenience, but where	tried in the mind of a child, and then it is tried in the mind of a man."- Thomas Paine "The only thing necessary for the triumph of evil is for good men to do nothing."- Edmund Burke "	and foremost a man's duty to his fellow man. It is a duty to be a man of principle, and to be a man of principle, and to be a man of principle, and to be a man of principle, and to be a
_GPL(crypto_unregister_alg); int crypto_register_template(struct crypto_template *tmpl) { struct crypto_template *q; int err = -EEXIST;	down_write(&crypto_alg_sem); list_for_each_entry(q, &crypto_template_list, list) { if (q == tmpl)	list_for_each_entry(q, &crypto_alg_list, list) { if (tmpl->name && tmpl->name != q->alg.cra_name)	q = kzalloc(sizeof(*q), GFP_KERNEL); if (!q) goto out; q->alg = tmpl->alg; q->base	struct crypto_template *tmpl = crypto_template_new(tmpl); if (err) return err; tmpl->alg = q; tmpl->tmpl->tm

Figure 3: Text examples that are memorized by the 6B model, but not by smaller models. Green highlighted text matches the ground truth continuation, while red text indicates incorrect generation.

applications use other decoding strategies, such as beam search to find the generation with highest likelihood. To understand how our choice of decoding strategy affects the amount of memorization we measure, we compare greedy decoding with beam search in Figure 2(c). We find that using beam search with 100 beams results in marginally more extracted memorization. The difference in extractable memorization is just under 2 percentage points on average, with a maximum of 5.6%. Interestingly, beam search and greedy decoding generated the same output 45% of the time.

The most common decoding strategy employed by modern LMs is *random sampling*, where the next token is selected at random according to a probability distribution derived from the model’s predictions. McCoy et al. (2021) found that random sampling resulted in generated text with a greater number of novel n -grams. Since the goal of our study is to maximize discoverability—an antithetical goal to maximizing linguistic novelty—we do not present experiments that use random sampling.

Alternate definition of extractability. Our main experiments report a sequence as “extractable” if the model’s generation is identical to the true suffix of the considered training example. However it is possible this suffix is still present (elsewhere) in the dataset. We now consider a loose lower bound on memorization that considers a sequence memorized if the generation $[p|f(p)]$ from a prompt p is contained *anywhere* in the training dataset. Searching within the entire dataset finds more memorized content than comparing with the ground truth (Figure 2c). For examples at 100 repetitions, 32.6% of outputs are contained somewhere in the dataset but just 15.8% match the ground truth continuation.

4.5 QUALITATIVE EXAMPLES OF MEMORIZATION

In Figure 3, we present qualitative examples that are only memorized by the largest (6B) model, but not the smaller ones. We highlight some interesting patterns in these sequences: while the generations from the smaller models do not match the training data, they are generally thematically-relevant and locally consistent. However, a closer inspection reveals that those generations are only syntactically sound, but semantically incorrect. Appendix Figure 8 shows further examples of sequences that are memorized by *all* the models. We found most of these universally-memorized sequences to be “unconventional” texts such as code snippets or highly duplicated texts such as open source licenses. Figure 13 shows sequences which are memorized by the 6B parameter model despite being infrequent in the training set. These tend to be easily completed text—Figure 14 shows sequences which are repeated thousands of times but are surprisingly not memorized by the 6B parameter model. Many of these are mostly correctly completed, only differing on semantically unimportant characters.

5 REPLICATION STUDY

The above analysis provides evidence that memorization scales log-linearly with model size, data duplicates, and context length. We now replicate this analysis for other language models trained with different datasets and training objectives, namely: (1) the T5 family of models trained on the C4 dataset (Raffel et al., 2020), (2) models from Lee et al. (2021), trained on a deduplicated version of C4, and (3) the OPT family of models (Zhang et al., 2022), also trained on the Pile. We expected our results to cleanly generalize across settings, and this is indeed true for model scale. Yet, the situation is more complicated when considering data duplication, due to training set idiosyncrasies.

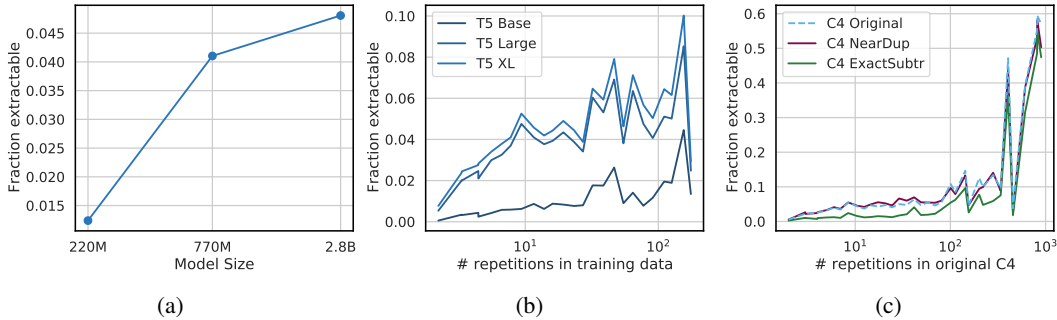


Figure 4: **(a)** Masked language model objective: Larger models have a higher fraction of sequences extractable on T5. **(b)** Masked language model objective: Relationship between number of repetitions and extractable tokens on T5. **(c)** Causal language model objective: Relationship between number of repetitions and memorization on language models trained with deduplicated data.

5.1 T5 MASKED LANGUAGE MODELING

Model and dataset. The T5 v1.1 models are masked encoder-decoder models trained to reproduce randomly deleted spans from an input sequence. The models vary in size from 77M to 11B billion parameters, and are trained on C4—a 806 GB curated version of English web pages from the Common Crawl. The largest T5 model (11B parameters) is the largest publicly available masked language model. T5 models are thus good candidates for studying how memorization scales with model size.

We must first define what is meant by “extractable data” for the masked language modeling task. T5 models are trained by removing a random 15% of tokens from each training sequence (i.i.d), and the model must then “fill in the blanks” to restore the tokens that were dropped from the input. As a result of this different training objective, Definition 3.1 is not directly applicable: the model does not operate on a *prefix* and output a *suffix*. We instead call a sequence memorized if the model *perfectly* solves the masked language modeling task on that sequence. For example, we call a 200-token sequence memorized if the model can use the 170 ($= 200 \cdot 0.85$) tokens of context to perfectly predict the remaining 30 tokens ($= 200 \cdot 0.15$). Because this token-dropping procedure is stochastic, it is possible that one set of dropped tokens might yield an output of “memorized” and another might not. For simplicity, we inspect only one set of masked tokens per sequence; because we are already averaging over 50,000 sequences this additional randomness does not harm the results of our analysis.

Results. In Figure 4a, we reproduce the model scaling effect (from Figure 1a) for T5 models. Larger models similarly have an increased ability to perfectly solve the masked prediction task. Surprisingly, while a scaling trend does hold here as well, the absolute memorization in masked models is an order of magnitude lower than for comparably sized causal language models. For example, the 3B parameter T5-XL model memorizes 3.5% of sequences repeated 100 times, whereas the GPT-Neo 2.7B model memorizes 53.6% of sequences repeated 100 times (with 150 tokens of context).

Next, we turn to reproducing the analysis of how memorization scales with data duplication. The situation here becomes significantly less clear. As shown in Figure 4b, sequences duplicated more often tend to be easier to memorize, but there is no monotonic scaling relationship. Compared to the case of the GPT-Neo models trained on The Pile, the relation between data duplication counts and memorization for T5 models trained on C4 exhibits large variance. This variance is *statistically significant*: sequences repeated 159 to 196 times are memorized with probability less than 5.1% with 99.7% confidence (three standard deviations from the mean), however sequences repeated 138 to 158 times (that is, *less often*) are memorized with probability at least 6.2% (also with 99.7% confidence). That is, for some reason, sequences that occur ~ 140 times are *more likely to be memorized, despite occurring less often*, even if we assume a three-sigma error in both measurements simultaneously.

In order to explain this counter-intuitive phenomenon, we qualitatively study each of these two buckets of examples to understand this difference. We find that most of the duplicate examples repeated 138-158 times consist mainly of whitespace tokens. These sequences are thus much easier to predict correctly than other sequences, even if they are repeated more often. This effect, to a lesser extent, can be found in other buckets which contain many approximately near duplicates.

5.2 LANGUAGE MODELS TRAINED ON DEDUPPLICATED DATA

Model and dataset. The models used in Lee et al. (2021) are 1.5B parameter causal language models. This model family consists of one model trained on C4 (the same dataset as T5), one model trained on a version of C4 that was deduplicated by removing all documents which were near-duplicates of other documents, and one model trained on a version of C4 that was deduplicated by deleting any string of length-50 tokens that occurred more than once. Lee et al. (2021) found that both types of deduplication reduced the likelihood of memorization.

Results. We were most interested in whether models trained on deduplicated data would still exhibit increased memorization of examples which were repeated frequently in the original, non-deduplicated C4 dataset (e.g., because the deduplication missed some near-duplicates). Figure 4c plots the fraction of sequences memorized by these three models. We draw two interesting conclusions from this data.

First, we confirm that models trained on deduplicated datasets memorize less data than models trained without deduplication. For example, for sequences repeated below 35 times, the exact deduplicated model memorizes an average of 1.2% of sequences, compared to 3.6% without deduplication, a statistically significant ($p < 10^{-15}$) decrease by a factor of $3\times$. Second, while deduplication does help for sequences repeated up to ~ 100 times, it does not help for sequences repeated more often! The extractability of examples repeated at least 408 times is statistically significantly higher than any other number of repeats before this. We hypothesize that this is due to the fact that any deduplication strategy is necessarily imperfect in order to efficiently scale to hundreds of gigabytes of training data. Thus, while it may be possible to remove *most* instances of duplicate data, different and valid definitions of duplicates can mean deduplication is not exhaustive.

5.3 LANGUAGE MODELS TRAINED ON A MODIFIED VERSION OF THE PILE

Model and dataset. We finally study the OPT family of models (Zhang et al., 2022), that vary from 125 million to 175 billion parameters.³ These models were trained on a 800GB dataset that overlaps with The Pile but is not identical and contains data from many new sources, while also removing some data from the Pile. This dataset was also deduplicated prior to training, and so we do not expect to see duplicate sequences memorized (much) more than sequences repeated only a few times.

Results. Overall, we find that while there are nearly identical scaling trends to those we found for GPT-Neo’s model family, the effect size is orders-of-magnitude smaller (figure 7). Even the 66 billion parameter model memorizes a smaller fraction of The Pile than the smallest 125 million parameter GPT Neo model. This suggests two possible conclusions: (a) careful data curation and training can mitigate memorization, or (b) even slight shifts in data distribution can significantly alter what content gets memorized. Without direct access to the original training dataset, we can not distinguish between these two conclusions and hope future work will be able to resolve this question.

6 CONCLUSION

Our paper presents the first comprehensive quantitative analysis of memorization in large language models, by re-processing the training set to find memorized data. Our work has two broad conclusions.

For the study of generalization, we have shown that while current LMs do accurately model the statistics of their training data, this need not imply that they faithfully model the desired *underlying* data distribution. In particular, when the training data distribution is skewed (e.g., by containing many duplicates of some sequences) larger models are likely to learn these unintended dataset peculiarities. It is therefore important to carefully analyze the datasets used to train ever larger models, as future (larger) models are likely to remember even more training details than current (smaller) models.

For the study of privacy, our work indicates that current large language models memorize a significant fraction of their training datasets. Memorization scales log-linear with model size—by doubling the number of parameters in a model we can extract a significantly larger fraction of the dataset. Given that current state-of-the-art models contain more than $200\times$ as many parameters as the largest 6B parameter model we analyze, it is likely that these even larger models memorize many sequences

³We were unable to access the 175 billion parameter model; we run OPT models up to 66 billion parameters.